Strong Thin Trees: Proof Architecture, Ram Coordinates, and Certified Local Extensions

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Abstract

The September 23, 2026 manuscript *The strong thin tree conjecture* claims a universal C/k bound for one spanning tree on every cut of every finite loopless k-edge-connected multigraph. We study its complete source at the pinned public snapshot, including the hierarchy, duality, geometric charging, fixed-point extraction, isotropic representation, interlacing, and iteration arguments. The directly relevant private report supplies fractional exactness and cut certificates, not the missing integral rounding theorem. We preserve that distinction and its correct mathematical content.

We rederive the extraction-to-iteration chain with its quantifiers, floors, and losses. We prove local extensions: residual-aware positive cut domination, physical-metric spectral certificates with coefficient and vector errors, a partition-specific packing certificate, and a sufficient criterion for approximate responses that exposes the margin hidden in a fixed-point limit. An elementary family separates cut from spectral thinness by an unbounded factor. Exact finite tests exercise these statements and selected source inequalities; they do not validate the universal hierarchy construction. Applicability level L5 denotes a direct match of mathematical targets, not a proof rating. The source's global conclusion is kept distinct from the proved local results, the external classical inputs, and the tests.

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1 Scope, status, and the exact mathematical target

The public object under study is OpenAI's manuscript [1], pinned at

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The review date is October 7, 2026. The main source is `build/source/main.tex`; all twenty TeX files and the bibliography are included in the reading ledger. The local download is a snapshot, not a Git checkout. Consequently provenance is checked against its SHA-256 manifest, not by claiming a successful local Git revision check.

For a finite loopless undirected multigraph $G = (V, E)$, parallel edges have separate identities. Write $\delta_G(S)$ for a nontrivial cut and $\lambda(G) = \min_S |\delta_G(S)|$. The target is

$$\exists C < \infty\ \forall k \geq 1\ \forall G\ (|V| \geq 2, \lambda(G) \geq k)\ \exists T \subseteq G\ \forall \varnothing \neq S \subsetneq V : \quad |\delta_T(S)| \leq \frac{C}{k}|\delta_G(S)|, \tag{1.1}$$

where T is a spanning tree and C is independent of $G, k, |V|$. This is stronger than Goddyn's unquantified connectivity function for a given thinness. Two vertices joined by k edges require $C \geq 1$.

We use four evidence classes throughout. *Source-claimed* denotes the manuscript's complete universal theorem. *Conditional global* denotes a deduction using the full stated hierarchy input. *Proved local* denotes results established here from explicit finite-dimensional or combinatorial premises. *Tested* means an executed finite check with its declared arithmetic, input range, and limitations. Reading every source file and finding no specific fatal gap is not a machine-checked global proof. Nor does an external catalogue declaration establish that its formal statement, hypotheses, and proof dependencies match the paper.

The private report TR-item-224-astra [2] addresses exactly the target in (1.1), hence the L5 applicability match. Its own internal research-level assessment is a different scale and is not overridden. Its September 7 literature assessment is historical; it must not be presented as an October 7 adjudication of a later preprint. The present report does not modify that private source or equate its certificates with a new universal rounding theorem.

2 Classical baseline and the private contribution

2.1 Fractional packing has the right order but the wrong quantifier

The Nash–Williams–Tutte criterion says that r edge-disjoint spanning trees exist exactly when every partition $\mathcal{P}$ satisfies $|E_G(\mathcal{P})| \geq r(|\mathcal{P}| - 1)$ [3, 4]. For a partition into $d \geq 2$ parts in a k-edge-connected graph,

$$|E_G(\mathcal{P})| = \frac{1}{2}\sum_{A \in \mathcal{P}} |\delta_G(A)| \geq \frac{kd}{2} \geq \lfloor k/2 \rfloor (d-1).$$

The one-part partition has zero demand and is treated separately. For $k \geq 2$, let $r = \lfloor k/2 \rfloor$ and choose packed trees T_i. Then $x = r^{-1}\sum_i \mathbf{1}_{T_i}$ is a spanning-tree-polytope point and

$$x_e \leq 1/r, \qquad x(\delta_G(S)) \leq |\delta_G(S)|/r \leq 2|\delta_G(S)|/(k-1). \tag{2.1}$$

This is one fractional vector satisfying all inequalities, equivalently an expectation bound under the packing mixture. It does not give an integral tree satisfying all of them, or even require the desired tree to be in that mixture's support.

The private report correctly separates marginal feasibility from negative dependence. Its K_5 paths $(01, 12, 23, 34)$ and $(02, 24, 41, 13)$ are edge-disjoint, but their uniform mixture has $\Pr(01, 12 \in T) = 1/2 > 1/4$. A different weighted-tree law can realize interior marginals by minimizing log-sum-exp minus the target linear form; boundary marginals follow by compact limits. Negative-dependence bounds are properties of those laws, not of an arbitrary mixture. This correction is preserved. The usual cut-counting union bound loses a logarithm: a cut of size ak has mean $O(a)$, while the number of such cuts can be $n^{O(a)}$. A tail of order $\exp[-\Theta(Ca \log C)]$ cannot beat that count with a universal C by this argument alone.

2.2 What the public literature establishes independently

Anari–Oveis Gharan's general result has a polylogarithmic-in-$\log n$ factor and supplies the locally connected hierarchy geometry [5]. Klein–Olver controls a prescribed laminar cut family [6]. Klein–Olver–Yeoh's 2026 theorem controls cuts strictly smaller than $(1 + 1/40)k$, with at most 88 tree crossings [7]. The strict threshold matters. These established restricted statements neither prove nor refute the later all-cut claim. Harvey–Olver provides the earlier spectral-halving framework [8]; MSS supplies a dimension-free independent-vector bound [9]. Their exact premises, rather than the names of the methods, are the relevant comparison.

2.3 Cut, spectral, and coarea formulations

Orient edges arbitrarily, put $b_{uv} = e_u - e_v$, and set $L_H = \sum_{e \in H} b_e b_e^T$. On $\mathbf{1}^\perp$,

$$\alpha_{\mathrm{cut}}(T; G) = \max_S \frac{\mathbf{1}_S^T L_T \mathbf{1}_S}{\mathbf{1}_S^T L_G \mathbf{1}_S}, \qquad \alpha_{\mathrm{spec}}(T; G) = \sup_{0 \neq z \perp \mathbf{1}} \frac{z^T L_T z}{z^T L_G z}. \tag{2.2}$$

Centering $\mathbf{1}_S$ does not change either Laplacian form, hence $\alpha_{\mathrm{cut}} \leq \alpha_{\mathrm{spec}}$. For weighted nonnegative graphs, the exact coarea identity is

$$\sum_{uv} w_{uv} |z_u - z_v| = \int_{\mathbb{R}} w(\delta\{v : z_v > t\})\, dt.$$

It proves equivalence of all-cut domination and domination of these ℓ^1 variations, not of the squared-difference energies. The private report's bound $\alpha_{\mathrm{spec}} \leq (n-1)\alpha_{\mathrm{cut}}$ follows by sorting z, writing its nonnegative successive gaps, and applying Cauchy–Schwarz on each edge's interval of gaps. We retain the bound without calling its factor sharp or dimension-free.

Proposition 2.1 (An explicit unbounded relaxation gap)**.** *Let G consist of a path $0, 1, \ldots, \ell$ with M parallel copies of each path edge, together with one edge 0ℓ, where $\ell \geq 2$ and $M \geq 2$. Let T consist of 0ℓ and one copy of every path edge except 01. Then*

$$\lambda(G) = M + 1, \qquad \alpha_{\mathrm{cut}}(T; G) = \frac{2}{M+1}, \qquad \alpha_{\mathrm{spec}}(T; G) \geq \frac{1 + (\ell-1)/\ell^2}{1 + M/\ell}.$$

Thus $M = \ell^2$ gives a spectral-to-cut ratio growing at least linearly in ℓ.

Proof. For a cut let p be the number of crossing edges of the underlying simple path, and let $c \in \{0, 1\}$ indicate whether 0ℓ crosses. Nontriviality gives $p \geq 1$, and parity gives $c \equiv p \pmod 2$. Its denominator is $Mp + c$, and its tree numerator is at most $p + c$. When $c = 0$ the ratio is at most $1/M \leq 2/(M+1)$; when $c = 1$ the inequality $(p+1)/(Mp+1) \leq 2/(M+1)$ is equivalent to $(M-1)(p-1) \geq 0$. A prefix cut crossing a retained path edge attains $2/(M+1)$. The smallest possible denominator is $M+1$, attained at a prefix cut; even p has denominator at least $2M$. Finally use $z_i = i/\ell$ and center it. Its graph energy is $M/\ell + 1$ and its tree energy is $(\ell-1)/\ell^2 + 1$. □

This example is a local diagnostic, not a counterexample to thin trees: it already exhibits a $2/\lambda(G)$-thin tree. It shows why the source needs cut-controlled shortcut matrices instead of silently replacing their upper bounds by $D \preceq L_G$.

3 The hierarchy mechanism and its parameter budget

3.1 The exact input required by extraction

A hierarchy is a rooted laminar tree of distinct vertex sets, rooted at V, with singleton leaves and children partitioning their parent. Local q-connectivity means $H[A]$ is q-edge-connected for every nonsingleton node A. Put $P(A) = \delta_H(A)$ and $O(A) = E_H(A, A^* \setminus A)$ for nonroot A. Each edge belongs to exactly two O-boundaries: the two children of the least node containing both endpoints. It can belong to many full P-boundaries. Substituting the latter for the former would destroy the multiplicity-two accounting.

The precise source input, `thm:hierarchy-power`, is: for sufficiently large absolute q, and any marked nodes satisfying $|O(A)| \geq |P(A)|/q$, there is a full-space $D \succ 0$ such that

$$\mathbf{1}_S^T D \mathbf{1}_S \leq |\delta_H(S)| \quad (\varnothing \neq S \subsetneq V), \qquad \frac{1}{|O(A)|} \sum_{e \in O(A)} b_e^T D^{-1} b_e \leq q^{-31/40}. \tag{3.1}$$

No constraint at $S = V$ is permitted, since $\mathbf{1}^T D \mathbf{1} > 0$ whereas the full cut is zero. No upper Loewner inequality against L_H is claimed. This full-space convention differs from the quotient-space Laplacian inverse used in the private spectral certificate.

3.2 Duality: weighted cuts are not silently replaced by arbitrary vectors

The appendix minimizes the maximum marked average over these D, with an infimum p_*. Slater feasibility follows from a small positive multiple of I, since the graph is connected and there are finitely many nontrivial cuts. Nonnegative multipliers λ_A, summing to one, weight the objective; nonnegative y_S weight only the cut constraints. Define

$$A = \sum_A \frac{\lambda_A}{|O(A)|} \sum_{e \in O(A)} b_e b_e^T, \quad Z = \sum_S y_S \mathbf{1}_S \mathbf{1}_S^T, \quad d(y) = \sum_S y_S |\delta_H(S)|.$$

The matrix minimization is exact even for singular A, Z:

$$\inf_{D \succ 0} \{\operatorname{tr}(AD^{-1}) + \operatorname{tr}(ZD)\} = 2 \operatorname{tr}[(Z^{1/2} A Z^{1/2})^{1/2}]. \tag{3.2}$$

For positive definite matrices, put $H = Z^{1/2} D Z^{1/2}$ and $C = Z^{1/2} A Z^{1/2}$. Subtracting $2 \operatorname{tr} \sqrt{C}$ gives $\|H^{-1/2} C^{1/2} - H^{1/2}\|_F^2$. Add εI to A, Z and pass to the limit for the singular case. Scaling y then optimizes $2\sqrt{s}\, N - sd$ to N^2/d.

Let B be the incidence matrix and $w_e^2 = \sum_{A: e \in O(A)} \lambda_A / |O(A)|$. The numerator is $\|Y B \operatorname{diag}(w)\|_*^2$ for rows $\sqrt{y_S} \mathbf{1}_S^T$ of Y. Approximate y by nonnegative rational weights and repeat indicator rows to obtain a binary embedding X. The common scale cancels from numerator and denominator. This is a supremum over arbitrary finite dimensions, not an exact finite representation of irrational weights.

Nuclear-norm duality introduces rows u_e of a matrix U with $UU^T = I_{|E|}$, after padding X with zero rows if needed. Row-sign changes allow $a_e = \langle u_e, X b_e \rangle \geq 0$. Multiplicity two gives $w_e \leq \sum_{A: e \in O(A)} \sqrt{\lambda_A / |O(A)|} \leq \sqrt{2} w_e$. Maximizing in λ yields

$$p_* \leq Q_* \leq 2p_*, \qquad Q_* := \sup_{X,U} \frac{\sum_A |O(A)|^{-1} (\sum_{e \in O(A)} a_e)^2}{\sum_e \|X b_e\|_2^2}. \tag{3.3}$$

The restriction $X \in \{0,1\}^{h\times V}$ is substantive: $\|Xb_e\|_2^2 = \|Xb_e\|_1$ permits geometric length charging. A generic real embedding would not have this identity. The proof uses no attainment of the original shortcut infimum: strict slack selects a feasible positive definite D after bounding Q_*.

3.3 Spectral packing and the construction of bags

The source's spectral-packing lemma has a transparent rank obstruction. For an m-edge set let A have columns given by restricted edge differences, let $\tau = m\sqrt{\Upsilon}$, and suppose $\sum \sigma_i \ge \tau$, $\sum \sigma_i^2 \le \tau^2/(am)$. Removing the first $\lfloor am/16 \rfloor$ singular values loses at most $\tau/4$. Dyadic index blocks therefore yield $p > am/16$ and

$$p\sum_{i\ge p}\sigma_i^2 \ge \frac{9\tau^2}{16L^2}, \qquad L \le 5 + \log_2(1/a).$$

At squared radius $r = (\sum_{i\ge p}\sigma_i^2)/(32m)$ a maximal separated endpoint set must contain at least $p+1$ points. Otherwise replacing endpoints by their nearby centers gives a matrix of rank at most $p-1$ with squared Frobenius error at most half that singular tail, contradicting the rank approximation inequality. Thus $br \ge 9m\Upsilon/(512L^2)$. The source weakens this to $a^\epsilon m\Upsilon/C_1(\epsilon)$ with $C_1 = 1024\epsilon^{-2}$, keeping the constants independent of m.

For an α-bad boundary, truncate small a_e and large d_e, where $d_e = \|Xb_e\|_1$. The surviving first moment is at least half the original. Two dyadic binnings in an interval of ratio $256/\alpha^2$ yield a 2-homogeneous subset whose squared boundary sum retains $\gamma = 1/(4J^4)$, with $J = 1 + \lceil \log_2(256/\alpha^2) \rceil$. Across homogeneous families, the source establishes $c_1 N' \ge sNc_2$, where $s = \alpha\gamma/32$. This density inequality supplies both a radius range and enough endpoint centers.

The resulting small balls either concentrate near relatively few larger balls, producing compact bags with many centers per diameter, or give many large balls that can be assigned to individual hierarchy nodes. Only centers incident to short edges of $O(t)$ are assigned to t. Grouping by scale and choosing one residue class modulo $H = 1 + \lceil \log_2(24/(\rho s^2)) \rceil$ makes successive radii geometrically separated. The other loss $L = 2 + \lceil \log_2(1/s) \rceil$ counts relative size bins. Neither is the number of hierarchy levels or the number of vertices.

3.4 Why geometric overlap does not automatically permit repeated charging

Compact bags are handled by genuinely disjoint ordinary balls and hollow shells in the continuous cube. A shell needs an inner embedded vertex and an outer embedded vertex. Integrating radial threshold cuts then charges at least k times its width. Along each fixed embedded edge segment, clamped radial functions are piecewise affine; disjoint shells permit at most one varying function at each interior point. Hence that edge pays at most its ℓ^1 length. This checks boundary segments and zero-length edges without treating an entire witness path as a new resource each time.

The compact construction maintains the nonnegative capacities $[w - 6\Delta_i]_+$ for old shells and $\delta_i - 6\Delta_{i+1}$ for fresh balls. If a bag has an interior ball, all its centers lie in a short radial interval; cutting that interval from the old shell costs at most $8\Delta_i$. Compactness $|D|\delta_i \ge 36\Delta_i$ pays the cost and preserves at least $|D|\delta_i/4$ new capacity. In the other case shrinking old shells by $2\delta_i + \Delta_i$ at each end makes room for remaining balls. The reserve inequality $2\Delta_i + 4\delta_i + 6\Delta_{i+1} \le 6\Delta_i$ transfers the capacity across phases. Positive parts prevent a negative-capacity accounting error when a shell disappears.

Assigned bags require more: their shells can overlap geometrically. Each record carries a connected rooted hierarchy domain with predecessor subtrees removed, and possibly one avoided descendant subtree. Records that overlap geometrically must have disjoint conflict domains. The

path condition requires at least $k/4$ edge-disjoint inner-to-outer paths in the allowed induced subgraph. Excluded predecessor boundary endpoints are outside the outer radius, so stopping a path at its first outer vertex removes all forbidden subtree visits. Consequently two records charging the same graph edge have intersecting domains and must be geometrically disjoint on that edge's fixed segment.

The preprocessing cost for a predecessor s is bounded by $2|P(s)|\sum_{i>\ell}\delta_i \le 4\lambda|P(s)|\delta_\ell$. Assignment and $|O(s)| \ge k\lambda|P(s)|$ convert this to at most $4/C_3$ of its bag mass. Deleting noninsertable balls and bags losing more than half their balls retains at least half the total mass.

The delicate operation creates two overlapping central shells. The first routes in $G[V(t)]$, the second in $G[V(t^*) \setminus V(t)]$. The latter graph is split into $k/4$-connected pieces; if there are j, local connectivity and the split cut budget imply $j \le 2|O(t)|/k$. This supplies many short-edge neighbors in one piece. The shells receive complementary conflict domains $C(B) \cap \mathcal{T}[t]$ and $C(B) \setminus \mathcal{T}[t]$. New path witnesses, rather than inherited geometry alone, establish their validity. The source explicitly rules out $t(B) = t$ at this operation; otherwise its strict-descendant avoidance label would be invalid. Its argument uses the fact that all remaining interior balls meeting a previously created central shell were already treated.

The three operations retain old allocations using avoiding-shell potential divided by $2(C_4+2)$ and nonavoiding potential without that divisor. At the phase end the shrink-cost inequality compares the loss in width to the gain from changing scale. With $C_4 = 3$, $C_3 = 10^4$, the final assigned-bag energy bound is

$$\mathcal{E} \ge \frac{q}{320000}\mathsf{M}. \tag{3.4}$$

These details are why a statement that "balls charge graph length" would not be an adequate verification of this appendix. The exact finite tests below check scalar budgets and counterexamples to shortcut substitutions, not every possible labeled arrangement.

3.5 Exponents and logarithms

The endpoint chooses

$$\alpha = q^{-4/5}, \quad \epsilon = 1/10, \quad \beta = 36, \quad \rho = (2q^2)^{-1}, \quad C_2 = 128J^4, \quad s = \alpha/C_2.$$

Then $C_2 = O((1+\log q)^4)$ and $L, H = O(1+\log q)$. The assignment condition is $s^{6/5} = q^{-24/25}C_2^{-6/5} \ge 24C_3/q$, since $q^{1/25}/C_2^{6/5} \to \infty$. The smallness condition $s^{1/10} \le (48\beta C_1)^{-1}$ also eventually holds. Good nodes contribute at most $2q^{-4/5}\mathcal{E}$; bad nodes are captured with mass at least $s^{1/10}\sum N_t/(384\beta C_1 C_2 LH)$. By (3.4),

$$Q_* \le 2q^{-4/5} + cq^{-23/25}(1+\log q)^7 = o(q^{-31/40}). \tag{3.5}$$

The unrounded logarithmic power is $32/5$, not a dependence on n. The positive exponent gaps are $1/40$ and $29/200$. Thus an absolute threshold exists. This derivation does not give a useful small numerical constant: the assignment margin grows only as $q^{1/25}$ against logarithms and large fixed constants. Testing moderately sized q would not test the regime in which the proof invokes this lemma.

4 Extraction and the fixed-point quantifiers

4.1 Maximal packable parts

Fix $q \ge 1$. Intersecting q-tree-packable induced vertex sets have a packable union: for each color contract the overlap in the second tree, choose a quotient tree, and attach it to the first tree. The

maximal sets therefore partition V and coarsen when edges are added. If their quotient has $d > 1$ vertices, it has fewer than $q(d-1)$ edges. Indeed, otherwise a smallest vertex set W with at least $q(|W|-1)$ internal edges has every proper subset below this density. Subtracting those internal bounds over a partition of W proves the Nash–Williams–Tutte inequalities there. Lifting its q quotient trees contradicts maximality. This is stronger than mere connected components and is the obstruction used at extraction's endpoint.

4.2 One response for every threshold

Let H pack r trees, $u = r^{-1/16}$, $q = \lfloor (1-u)r \rfloor$. The compact convex domain is

$$\mathcal{K} = \{X \succeq L_H + n^{-1}I : \mathbf{1}_S^T X \mathbf{1}_S \le 3|\delta_H(S)| \text{ for all proper nonempty } S\}.$$

It is nonempty since $|S|/n < 1 \le |\delta_H(S)|$. Singleton cuts bound the diagonal; positivity bounds all off-diagonal entries, hence the set is compact. Every inverse has an eigenvalue margin $1/n$. This graph-dependent margin is acceptable for existence, but not a uniform bit-complexity bound.

At a fixed X, threshold its resistances to form $F_s(X)$ and let $\mathcal{P}_s(X)$ be its maximal q-packable partition. The crossing sets $B_s = E_H(\mathcal{P}_s)$ are nested as s increases. Choose the finest representative partition in each occupied dyadic bin of $M_s = |B_s|$, and append the root partition. Their distinct parts form a locally q-connected hierarchy. At each selected transition the disappearing edges lie in the parent boundaries of changing parts, and the total full-boundary size of those parts is at most $2M_j$. Unmarked parts cost at most $2M_j/q$; marked ones cost at most $2r^{-1/4}M_j$, after (3.1) and Markov counting. The exponent conversion is $q^{-31/40} \le 2^{31/40} r^{-31/40} \le r^{-3/4}$ for $r \ge 2^{31}$.

Since $\sum_{j \ge i} M_j < 4M_i$ and the representative in the same bin has $M_i < 2M_s$, one matrix $Y = L_H + n^{-1}I + D \in \mathcal{K}$ satisfies

$$|\{e \in B_s(X) : R_Y(e) > r^{-1/2}\}| \le 40 r^{-1/4} |B_s(X)| \qquad \text{for every } s > 0. \tag{4.1}$$

The same Y works at all thresholds. Neither choosing a different matrix after each cut nor adding a loss for every hierarchy level is part of this deduction.

4.3 Why discontinuity is not ignored

The response need not be continuous. The source covers $\mathcal{K}$ by finitely many open balls of radius $1/p$, chooses responses at their centers, and forms a continuous convex combination using tent weights $(p^{-1} - \|Z - C\|)_+$. Brouwer gives a fixed point X_p of this interpolated map. Compactness gives a subsequential limit X. The fixed-point appendix proves Brouwer through shrinking barycentric subdivisions, Sperner parity, and nearest-point projection from an enclosing simplex. The projection acts in the affine hull, so a lower-dimensional $\mathcal{K}$ creates no hidden interior assumption.

Choose $s \in (3r^{-1/2}, 4r^{-1/2})$ distinct from the finitely many $R_X(e)$. For large p, every active center has exactly $F_s(X)$. Thus its bad-edge count in (4.1) concerns the same finite set B. Weighted counting implies all but at most $80r^{-1/4}|B|$ edges are good in at least half the response weight. For such an edge, $R_Y(e) \le r^{-1/2}$ is equivalent to $Y \succeq r^{1/2} b_e b_e^T$, whence $X_p \succeq \frac{1}{2} r^{1/2} b_e b_e^T$. The condition that at least N edges satisfy this is a finite union of closed intersections, so it passes to X. This last step is necessary: the identities of the good edges may change with p.

If the stabilized partition had $d > 1$ parts, the original packing would give $|B| \ge r(d-1)$. For sufficiently large r, $80r^{-1/4} < u$, so the threshold graph has more than $q(d-1)$ crossing edges, contradicting maximality. Consequently it packs q trees, each edge satisfying $R_X(e) < 4r^{-1/2}$. This is a valid conditional deduction from the full hierarchy theorem; it does not use an averaging theorem for trees. The finite-cover resolution remains graph-dependent.

5 MSS, rank preservation, and the final all-cut tree

5.1 A positive representation of the packed base family

Let B be exactly the union of q edge-disjoint spanning trees. In orthonormal coordinates on $\mathbf{1}^\perp$, set

$$N(a) = \sum_{e\in B} e^{a_e} b_e b_e^T, \qquad h(a) = \log\det N(a) - \frac{1}{q}\sum_{e\in B} a_e.$$

Cauchy–Binet contains each packed tree determinant as a positive summand. Arithmetic–geometric mean over those q summands bounds h below by a constant L. Minimize $h(a) + \gamma\|a\|^2$. At a minimizer, comparison with zero gives $\gamma\|a\|^2 \le h(0) - L =: K$ and $\|\nabla h(a)\| = 2\gamma\|a\| \le 2\sqrt{\gamma K}$. For any $\rho > 0$ choose γ small enough that every derivative has absolute value at most ρ/q. Then

$$z_e = e^{a_e/2} N(a)^{-1/2} b_e, \qquad \sum_{e\in B} z_e z_e^T = I, \qquad \|z_e\|^2 \le (1+\rho)/q. \tag{5.1}$$

Every weight is strictly positive and the common linear map is invertible. Thus every edge subset retains its incidence rank. An exact minimizer of the unregularized objective is unnecessary; permitting zero edge weights would lose the rank premise.

If $A \subseteq B$ satisfies $\sum_{e\in A} z_e z_e^T \succeq cI$, let W span all within-part columns for a partition into d parts. Then $\dim W \le n - d$, so projection P onto $W^\perp$ has trace at least $d - 1$. Taking a projected trace gives

$$c(d-1) \le \sum_{e\in E_A(\mathcal{P})} \|Pz_e\|^2 \le \frac{1+\rho}{q}|E_A(\mathcal{P})|. \tag{5.2}$$

Hence A packs at least $\lfloor cq/(1+\rho)\rfloor$ trees. This is a certificate about unit-multiplicity edge identities, not weighted connectivity. One edge of weight q is not a replacement for q separate edges when counting edge-disjoint spanning trees.

Proposition 5.1 (Partition-specific packing certificate)**.** *Let the z_e be positive rescalings of incidence columns under one invertible map, for a connected graph B. For each nontrivial partition $\mathcal{P}$, let $P_\mathcal{P}$ project onto the orthogonal complement of the span of its within-part columns, and set*

$$m_\mathcal{P} = \max_{e\in E_B(\mathcal{P})} \|P_\mathcal{P} z_e\|^2.$$

For $A \subseteq B$, put $S_A = \sum_{e\in A} z_e z_e^T$. If an integer $r \ge 1$ satisfies, for every such partition,

$$\operatorname{tr}(P_\mathcal{P} S_A) \ge r(|\mathcal{P}| - 1) m_\mathcal{P},$$

then A packs r edge-disjoint spanning trees. It suffices to replace S_A on the left by any proved lower bound $D_A \preceq S_A$.

Proof. Connectedness and rank preservation give $m_\mathcal{P} > 0$: otherwise all columns would lie in a subspace of dimension at most $n - |\mathcal{P}| < n - 1$. Within-part columns project to zero, so

$$\operatorname{tr}(P_\mathcal{P} S_A) = \sum_{e\in E_A(\mathcal{P})} \|P_\mathcal{P} z_e\|^2 \le |E_A(\mathcal{P})| m_\mathcal{P}.$$

The premise therefore implies every partition inequality for r packed trees. A lower bound D_A can only decrease the projected trace. □

This extends the uniform bound (5.2): it permits partition-dependent projected norms and lower traces, without requiring one worst eigenvalue to pay for every partition. It is a sufficient certificate, not a polynomial-time procedure; the number of partitions can be exponential. A physical-metric congruence is applied before forming these Euclidean projections, so arbitrary coefficient scaling cannot improve the condition spuriously.

5.2 Independent vectors, not independent tree edges

The MSS theorem applies to independent finitely supported random vectors with covariance sum I and expected squared norms at most ε, giving an outcome with sum of outer products at most $(1+\sqrt{\varepsilon})^2 I$. The paper's appendix recovers this via stable determinant polynomials, the expectation identity, common interlacing, and barrier shifts. Independence is used in the conditional determinant identity. Common interlacing is justified because every convex combination of the children is real-rooted, not merely because each child is real-rooted.

The barrier argument is internally explicit. Above the roots, the rational function $\Phi_p^i(z+te_j)$ has nonnegative constant term and positive simple residues at negative real poles. The upper half-plane sign of its imaginary part excludes higher-order poles and polynomial terms of degree at least two; nonnegativity on the positive real axis excludes a negative linear term. Consequently the barrier is decreasing and convex. For $\Phi_p^j(z) + 1/\delta \le 1$, the identity

$$\Phi^i_{(1-\partial_j)p}(z+\delta e_j) = f(\delta) - \frac{f'(\delta)}{1-\Phi_p^j(z+\delta e_j)} \le f(0)$$

uses both signs correctly. Starting at $t = \varepsilon + \sqrt{\varepsilon}$ and shifting each coordinate by $\delta = 1 + \sqrt{\varepsilon}$ gives the root bound $t + \delta = (1+\sqrt{\varepsilon})^2$. This matches the independently published theorem; no tree-law negative dependence is substituted.

Lemma 5.2 (Simultaneous two-sided splitting)**.** *If $\Sigma = \sum_e w_e w_e^T \preceq cI$ and $\|w_e\|^2 \le \varepsilon$, with $c, \varepsilon > 0$, there is A such that*

$$\tfrac{1}{2}\Sigma - \tau_c I \preceq \sum_{e\in A} w_e w_e^T \preceq \tfrac{1}{2}\Sigma + \tau_c I, \qquad \tau_c = \sqrt{2c\varepsilon} + \varepsilon.$$

Proof. First take $c = 2$. Independently place w_e in the first or second block of $\mathbb{R}^{2D}$ with equal probabilities. Its covariance sum is $M = \operatorname{diag}(\Sigma/2, \Sigma/2) \preceq I$. Split each positive eigenvalue λ of $I - M$ into $\lceil \lambda/\varepsilon \rceil$ equal rank-one deterministic vectors. Their squared norms are at most ε, and they make the covariance exactly isotropic. Apply MSS and subtract this fixed padding. The two diagonal blocks give the upper bound on A and on its complement, hence both sides with $\tau = 2\sqrt{\varepsilon} + \varepsilon$. For general c, apply this result to $\sqrt{2/c}\, w_e$ and multiply by $c/2$. □

Singular Σ causes no problem. Padding is deterministic and independent, and it is subtracted only after obtaining the full matrix inequality.

5.3 The same selected edges obey both demands

Take the extracted X, set $a_e = X^{-1/2} b_e$ for $e \in B$, and stack $w_e = (a_e, z_e)$ with $\rho = u$. Their covariance has generally nonzero cross blocks. Nevertheless

$$\sum_e (\langle x, a_e\rangle + \langle y, z_e\rangle)^2 \le 2\|x\|^2 + 2\|y\|^2,$$

since $L_B \preceq L_H \preceq X$ and $\sum z_e z_e^T = I$. Also $\|w_e\|^2 \le 4r^{-1/2} + (1+u)/q \le 6r^{-1/2}$ for the declared large-r regime. Lemma 5.2 with $\varepsilon = 6r^{-1/2}$ yields on the same A

$$\sum_{e\in A} z_e z_e^T \succeq (1/2 - \tau) I, \qquad L_A \preceq \tfrac{1}{2} L_H + \tau X, \qquad \tau = 2\sqrt{6}\, u^4 + 6u^8. \tag{5.3}$$

Testing only the second inequality on cut indicators gives $|\delta_A(S)| \le (1/2 + 3\tau)|\delta_H(S)|$. The first gives a packing, using (5.2). No graph-independent inequality $X \preceq 3L_H$ is invoked.

For $u \le 1/16$, $\tau \le u/2$. The source retains the floor loss through

$$(1-2\tau)((1-u)r-1) - (1+u)(1-4u)r \ge r(u+5u^2) - (1-u) \ge 0.$$

Thus $r' = \lfloor (1-4u)r/2 \rfloor$ packed trees remain. Since $1/r \le u$,

$$5/16 \le r'/r \le 1/2, \qquad \frac{1/2+3\tau}{r'/r} \le \frac{1+3u}{1-6u} \le 1+15u. \tag{5.4}$$

The lower bound on r includes 16^{16} and the hierarchy and extraction thresholds. The numerics in this report do not claim to construct graphs with that many packed trees.

5.4 Summable loss and the final quantifier

For $r_i \ge r_*$ apply the same-subset step and stop when $1 \le r_j < r_*$. Since $r_{i+1} \le r_i/2$, going backwards gives $r_i \ge 2^{j-1-i} r_*$ for $i < j$. Therefore

$$\sum_{i<j} r_i^{-1/16} \le \frac{r_*^{-1/16}}{1-2^{-1/16}}, \qquad C_{\mathrm{it}} := \exp\left(\frac{15 r_*^{-1/16}}{1-2^{-1/16}}\right).$$

Multiplying (5.4) telescopes the packing counts:

$$|\delta_{H_j}(S)| \le C_{\mathrm{it}} \frac{r_j}{r_0} |\delta_{H_0}(S)|.$$

This holds for every cut of one nested sequence of subgraphs. Any spanning tree of H_j inherits it. The zero-round case has an empty product and is valid because $C_{\mathrm{it}} r_*/r_0 \ge 1$. Finally $r_0 = \max(1, \lfloor k/2 \rfloor) \ge k/3$ gives $C = 3C_{\mathrm{it}} r_*$. This proves the global deduction conditional on (3.1); it is not a new independent proof of that input.

Proposition 5.3 (General loss criterion)**.** *Suppose each step has $0 < r_{i+1} \le \theta r_i$, $\theta < 1$, and all-cut factor at most $(r_{i+1}/r_i)(1 + A r_i^{-a})$, where $a > 0$. Stop below R while retaining positive packing. Then the accumulated relative loss is at most $\exp(AR^{-a}/(1-\theta^a))$, independently of the number of steps.*

Proof. Backward geometric growth gives $r_i \ge \theta^{-(j-1-i)} R$. Sum the resulting geometric series of r_i^{-a} and use $\log(1+x) \le x$ in the product. The packing ratios telescope. □

More generally the sufficient condition is a uniform bound on the sum of actual step errors. A per-step constant error would grow with the number of rounds and cannot replace the decay above.

6 Ram coordinates and rigorous local certificates

6.1 Preserve the useful private mathematics and correct unsafe transfers

TR190's spectral branch defines $K_j = \kappa_j(L_G)$. If $g(\lambda) = \sum_j |\kappa_j(\lambda)|^2$ lies between positive bounds on the retained spectrum, then $\sum_j K_j^* K_j = g(L_G)$ and $g(L_G)^{-1} \sum_j K_j^* K_j = I$ there. This is ordinary finite spectral functional calculus, with an exact Ram coefficient interpretation. TR248 supplies physical-model frame margins and perturbation budgets; TR250 supplies simultaneous transport of incidence maps and physical forms [10, 11, 12].

Two broader statements in TR190 cannot be imported unchanged. An ordering cannot make an arbitrary local matrix unit triangular: a permutation similarity preserves eigenvalues, whereas

$\begin{pmatrix} 1 & a \\ a & 1 \end{pmatrix}$ has eigenvalues $1 \pm a$, not both 1, for $a \neq 0$. Nilpotence must be an actual hypothesis for the finite Neumann inverse. Likewise small heat scale is not exact finite-hop support. On a connected graph choose $c \geq \max_v \deg(v)$; $A = cI - L_G$ is nonnegative. For any two vertices a path gives $(A^r)_{uv} > 0$ for some r, so $(e^{-tL_G})_{uv} = e^{-tc}(e^{tA})_{uv} > 0$ for every $t > 0$. By contrast a polynomial of degree r in L_G has at most r-hop support. These proofs preserve the correct filter and frame content while excluding unsupported triangularity and locality.

6.2 Complete parity coordinates for a candidate tree

Root a candidate tree T. For every graph edge let p_e be its tree-path mask in $\mathbb{F}_2^{n-1}$. Each nonzero mask F defines a vertex coloring by parity along root paths. Adjacent tree vertices differ exactly on F, and graph-edge endpoints differ exactly when $\langle p_e, F\rangle = 1$. Thus

$$|\delta_T(S_F)| = |F|, \quad |\delta_G(S_F)| = \sum_e \mathbf{1}_{\langle p_e, F\rangle = 1}. \tag{6.1}$$

These are all cuts up to complementation, because choosing the root color fixes every vertex color. If $h(p)$ counts edges of each mask, $\widehat{h}(F) = \sum_p h(p)(-1)^{\langle p,F\rangle}$ gives denominator $(|E| - \widehat{h}(F))/2$. A Walsh transform is a complete finite certificate at cost $O((n-1)2^{n-1})$ after path preprocessing, not a polynomial-time method on general graphs. Restricting $|F| \leq q$ produces only a lower bound on the full maximum unless a coverage theorem is supplied. This rederives and preserves the private result.

Theorem 6.1 (Residual-aware positive domination)**.** *Let $c(S)$ be the graph-edge cut incidence vector and let T be one fixed tree. Suppose $|\delta_T(A)| \leq \alpha_A |\delta_G(A)|$ on a protected family. For each cut S suppose there are nonnegative coefficients a_A and a nonnegative vector h_S satisfying*

$$c(S) \leq \sum_A a_A c(A) + h_S.$$

Then

$$|\delta_T(S)| \leq \sum_A a_A \alpha_A |\delta_G(A)| + \sum_{e \in T} h_{S,e}. \tag{6.2}$$

In particular, if $\alpha_A \leq C/k$, the protected budget is at most $B|\delta_G(S)|$, and the residual tree load is at most $\eta|\delta_G(S)|/k$, then T is $(BC + \eta)/k$-thin.

Proof. Pair the componentwise domination with the nonnegative incidence vector of T, then apply the protected bounds term by term. The stated budgets give the last conclusion. All sums are finite. □

The private theorem is the case $h_S = 0$, $\alpha_A = C/k$. A known residual support of at most b tree edges with $h_{S,e} \leq 1$ gives additive b, which is absorbed as $b|\delta_G(S)|/k$ when $\lambda(G) \geq k$. Obtaining a uniform constant residual is thus already as strong as the target, not a harmless weakening. Signed uncrossing identities need additional sign control before they can serve as the nonnegative domination premise.

6.3 Congruence, the constant mode, and physical errors

Let Q have orthonormal columns spanning $\mathbf{1}^\perp$. A square invertible Ram synthesis R defines $J = QR$ and

$$M = J^T J = R^T R, \quad A = J^T L_T J, \quad B = J^T L_G J. \tag{6.3}$$

The coefficient of the centered cut is $c_S = R^{-1}Q^T\mathbf{1}_S$. It satisfies $c_S^T A c_S = |\delta_T(S)|$ and $c_S^T B c_S = |\delta_G(S)|$. Therefore invertible coordinates preserve both the generalized spectral quotient and the discrete set of cut quotients when that set is transported too. Replacing c_S by arbitrary binary coefficient vectors changes the problem. For shortcuts positive definite on the full space, use a full invertible synthesis and retain the constant coordinate; projection onto $\mathbf{1}^\perp$ alone is not a substitute for their cut contract.

Theorem 6.2 (Physical-metric error certificate)**.** *Let $M \succ 0$, $B \succeq \gamma M$ with $\gamma > 0$, and suppose*

$$-e_A M \preceq A - \widetilde{A} \preceq e_A M, \quad -e_B M \preceq B - \widetilde{B} \preceq e_B M, \quad \widetilde{A} \preceq \widetilde{\alpha}\widetilde{B}, \quad \widetilde{\alpha} \geq 0.$$

Then

$$A \preceq \left(\widetilde{\alpha} + \frac{e_A + \widetilde{\alpha} e_B}{\gamma}\right) B. \tag{6.4}$$

If instead one knows $\widetilde{B} \succeq \widetilde{\gamma} M$ with $\widetilde{\gamma} > e_B$, one may set $\gamma = \widetilde{\gamma} - e_B$.

Proof. The premises imply $A \preceq \widetilde{\alpha} B + (e_A + \widetilde{\alpha} e_B) M$. Substitute $M \preceq B/\gamma$. The alternative lower bound follows by subtracting $e_B M$. □

These errors are norms after congruence by $M^{-1/2}$; unscaled entrywise errors need not be small in that metric. For rational matrices, exact LDL^T factors with positive diagonal pivots give independently checkable strict certificates. A semidefinite endpoint needs a rank-aware factorization or a proved limiting argument.

Proposition 6.3 (Errors in the cut coordinate itself)**.** *Suppose $A \succeq 0$, $B \succeq 0$, $A \preceq L_A M$, $B \preceq L_B M$, and $c = \widehat{c} + d$ with $\|d\|_M \leq \zeta$. Put $E = 2\zeta\|\widehat{c}\|_M + \zeta^2$. Then*

$$c^T A c \leq \widehat{c}^T \widetilde{A} \widehat{c} + e_A \|\widehat{c}\|_M^2 + L_A E, \quad c^T B c \geq \widehat{c}^T \widetilde{B} \widehat{c} - e_B \|\widehat{c}\|_M^2 - L_B E.$$

Whenever the latter bound is positive, the ratio of these two bounds is an upper certificate for that exact cut quotient.

Proof. Expand the quadratic form at $\widehat{c}$. The mixed term has absolute value at most $2L\|\widehat{c}\|_M\zeta$, and the remaining term at most $L\zeta^2$, by the operator bound in the physical metric. Apply the matrix approximation errors separately at $\widehat{c}$. □

This is useful for numerical coordinates but not a license to infer exact integer cut counts from noisy graph observations. The graph and its multiplicities remain exact input data.

6.4 Concrete graph and coordinate certificate

For $G = K_4$ and the path $T = (01, 12, 23)$, direct cut enumeration gives $\alpha_{\text{cut}} = 3/4$, attained by the alternating cut $\{1, 3\}$. The full spectral value is $(2+\sqrt{2})/4 < 7/8$, from the path Laplacian eigenvalues on $\mathbf{1}^\perp$. A rational certificate avoids eigenvalue numerics: take columns $e_0 - e_3, e_1 - e_3, e_2 - e_3$ in J_0 and

$$R = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 1/3 \end{pmatrix}, \quad J = J_0 R, \quad M = J^T J, \quad A = J^T L_T J, \quad B = 4M.$$

The checker verifies exact positive LDL^T pivots of $7B/8 - A$, all seven cut coordinate identities, and the transformed incidence rank. If $\widetilde{A} = A - M/100$ and $\widetilde{B} = B + M/200$, then $\widetilde{A} \preceq (7/8)\widetilde{B}$ and Theorem 6.2 yields the explicit bound

$$\alpha_{\text{spec}} \leq \frac{7}{8} + \frac{1/100 + (7/8)/200}{4} = \frac{5623}{6400}.$$

The exact 7/8 certificate is sharper; the perturbed example tests the correctness of a complete error budget, not an improvement in the physical thinness through coordinate scaling.

6.5 Explicit extraction cutoff and a residual criterion

The following refinement makes the dependence on the hierarchy premise visible. Let $q_H \geq 2$ be its absolute integer cutoff and put

$$R_0 = \max\{2^{34}, 2q_H\}, \qquad C_0 = 40.$$

For $r \geq R_0$, set $u = r^{-1/16}$, $t = r^{-1/2}$, $q = \lfloor (1-u)r \rfloor$, and $\delta = 40r^{-1/4}$. Then $q \geq r/2 \geq q_H$, the hierarchy-to-response exponent comparison holds because $r \geq 2^{31}$, and $2\delta < u$. The last assertion follows from $r^3 > 80^{16}$, which holds already at $r = 2^{34}$. Thus this cutoff is explicit *relative to* q_H, not an explicit numerical bound for the entire source theorem. The following statement also identifies what an approximate fixed point would actually have to certify. It incorporates the scoped mathematical check recorded in the confidential extraction audit [14].

Theorem 6.4 (Residual extraction criterion, conditional on responses)**.** *Let H, r, q, t, δ be as above, and let K be the compact extraction domain. Suppose $Z, C_j, Y_j \in K$, $\lambda_j \geq 0$, $\sum_j \lambda_j = 1$, and each Y_j is the simultaneous hierarchy response to C_j. Write $A = \sum_j \lambda_j Y_j$. Choose $3t < s < 4t$ with $\eta = \min_{e \in E(H)} |R_Z(e) - s| > 0$. Suppose every active index satisfies*

$$\|C_j - Z\| < \frac{\eta}{2n^2}, \qquad \|A - Z\| \leq \varepsilon \leq \frac{1}{2n}.$$

Then the graph of edges with $R_Z(e) \leq s$ contains q edge-disjoint spanning trees. In particular, Z meets the extraction conclusion.

Proof. For $U, V \in K$ the inverse identity and $U, V \succeq n^{-1}I$ give

$$\|U^{-1} - V^{-1}\| \leq n^2 \|U - V\|, \qquad |R_U(e) - R_V(e)| \leq 2n^2 \|U - V\|.$$

Hence all active centers have the same threshold graph at s as Z, and the same maximal packing partition and crossing set B. Their responses each have at most $\delta|B|$ bad edges at threshold t. Averaging bad indicators shows that all but at most $2\delta|B|$ edges have good-response weight at least $1/2$. Rank-one order therefore gives $A \succeq b_e b_e^T/(2t)$ for these edges, hence $R_A(e) \leq 2t$. Since $A \preceq Z + \varepsilon I \preceq (1 + n\varepsilon)Z$, inverse order gives

$$R_Z(e) \leq (1 + n\varepsilon) R_A(e) \leq 3t < s.$$

If the common partition has $d > 1$ parts, its H crossing size is at least $r(d-1)$. It consequently has more than $(1-u)r(d-1) \geq q(d-1)$ edges of the threshold graph, contradicting the strict maximal-packing-partition bound. For $B = \varnothing$, connectedness already forces $d = 1$. □

No matrices, response list, threshold margin, or residual satisfying this theorem have been computed here. The finite checker verifies the cutoff arithmetic and elementary response inequalities, not an instance of this certificate or the existence of computable hierarchy responses.

Proposition 6.5 (Averaged quadratic-response error)**.** *Let $X \succeq \mu I > 0$, $A = \sum_j \lambda_j D_j$, and $\|A - X\| \leq \varepsilon$. For any nonzero vector b, define*

$$h_j = \frac{b^T X^{-1} D_j X^{-1} b}{b^T X^{-1} b}.$$

Then $\left|\sum_j \lambda_j h_j - 1\right| \leq \varepsilon/\mu$.

Proof. With $v = X^{-1}b$, the absolute numerator error is at most $\varepsilon\|v\|^2$, whereas $b^T X^{-1} b = v^T X v \geq \mu\|v\|^2$. Division proves the claim. □

This identity is sometimes useful for checking a proposed response average. It is not an identity for averaged inverse matrices and does not replace the rank-one argument in Theorem 6.4.

6.6 Approximate resistance thresholds that preserve packing

Proposition 6.6 (Outer-threshold transfer)**.** *Suppose $X, \widehat{X} \succ 0$ on the same space and $(1-\varepsilon)X \preceq \widehat{X} \preceq (1+\varepsilon)X$, where $0 \leq \varepsilon < 1$. If the edges with $R_X(e) \leq s$ contain q packed trees, then*

$$E_{\text{out}} = \{e : R_{\widehat{X}}(e) \leq s/(1-\varepsilon)\}$$

contains q packed trees, and every edge of this set has $R_X(e) \leq s(1+\varepsilon)/(1-\varepsilon)$. If validated intervals $[\ell_e, u_e]$ enclose $R_{\widehat{X}}(e)$ and have width at most h, selection by $\ell_e \leq s/(1-\varepsilon)$ still preserves packing and gives

$$R_X(e) \leq (1+\varepsilon)\big(s/(1-\varepsilon) + h\big).$$

Proof. Congruence, inversion, and evaluation on b_e give

$$\frac{R_X(e)}{1+\varepsilon} \leq R_{\widehat{X}}(e) \leq \frac{R_X(e)}{1-\varepsilon}.$$

The outer set therefore includes every original core edge. Its upper true-resistance bound follows from the left inequality. For intervals, every core edge has $\ell_e \leq R_{\widehat{X}}(e) \leq s/(1-\varepsilon)$; every selected edge has $R_{\widehat{X}}(e) \leq u_e \leq \ell_e + h$. Combine these observations with the same left inequality. □

The smaller threshold $s/(1+\varepsilon)$ certifies true low resistance but may lose packing: even one essential boundary edge with $R_X(e) = s$ and $\widehat{X} = (1-\varepsilon)X$ is discarded when $\varepsilon > 0$. Similarly, selection by an interval's upper endpoint does not automatically retain the true core. If X has the extraction cut bound, $\widehat{X}$ has the bound multiplied by $1+\varepsilon$. This lemma, also recorded in the supplemental transfer note [15], starts from an established extraction certificate and validated relative error. It neither constructs the fixed point nor asserts a precision or runtime bound. It is distinct from Theorem 6.4, which supplies sufficient conditions for a proposed approximate response itself.

7 Assessment and research consequences

7.1 Strengths and limits of the existence architecture

The source separates three genuinely different obstructions. The hierarchy controls averaged resistances with only cut upper bounds; extraction turns that information into one packed low-resistance core; the positive isotropic representation and one simultaneous MSS selection preserve both spectral control and unweighted packing. The final iteration has summable losses. In particular, there is no unaccounted union bound over all cuts, no separate tree chosen for each cut, and no assumption that a spectral lower bound alone preserves a prescribed number of unweighted trees.

The price is substantial. The hierarchy cutoff is absolute but very large and not numerically instantiated here. Its assigned-shell conflict domains are the most intricate combinatorial component. Extraction uses compactness and a graph-dependent threshold gap; the proof does not give a practical finite cover. Isotropic normalization uses a regularized minimizer with strictly positive, potentially poorly conditioned weights. MSS gives an existence choice, and the all-cut guarantee

remains tied to the original graph through every round. These limitations concern quantitative usefulness and certification, not a counterexample to the stated theorem.

The reading and rederivations did not identify a specific fatal defect in the pinned manuscript. This is an analytical assessment, not certification that every labeled-shell configuration or every imported formal dependency has been mechanically validated. The global conclusion is the source's claim; the extraction and iteration deductions in this report remain explicitly conditional on the complete hierarchy premise. None of the local Ram transformations removes that premise or proves a stronger universal thin-tree constant by a change of coordinates.

7.2 A current primary-literature boundary

The public refresh also found Kathuria's September 2026 preprint, version 2, whose Theorem 10.2 claims one tree simultaneously $O(\varepsilon \log(2s))$-spectrally thin for s positive weightings on a common support, assuming every edge leverage is at most ε in each weighting [13]. Its stated reduction retains original edge weights and uses a higher-rank signing result. The common support and uniform leverage hypotheses are essential; this is not an all-cut theorem from edge connectivity alone. Only the theorem, its reduction, and the discussion distinguishing these hypotheses were inspected for this refresh. The signing proof and its claimed formalization were not independently verified. Version 2 states a single logarithm; a version-1 squared-logarithm description should not be carried forward as the current statement.

7.3 Focused next steps

The highest-value next work is to extract checkable finite witnesses for the actual hierarchy operation, including conflict domains, avoided subtrees, path counts, and width allocations. Exact rational tests of scalar inequalities cannot replace those witnesses. A second direction is to make the absolute hierarchy cutoff explicit from all simultaneous inequalities, even if the resulting number is enormous. This would make the final constant a reproducible expression with no hidden "large enough" dependence.

For computational certification, retain exact graph edge identities and use interval or rational arithmetic for the matrix layer. The residual criterion requires a threshold margin and a response residual; the outer threshold lemma permits controlled resistance inflation without deleting the packed core. Neither should be advertised as an algorithm until the response construction, precision demand, and full runtime model are supplied. Ram frames can then serve as an exact coordinate system or a conditioning device with transported metrics and error budgets. Restricting to a small family of Ram cut modes needs a proved positive-domination or coverage theorem before it controls all cuts of one tree.

8 Reading and evidence ledger

8.1 Pinned manuscript: complete mathematical-source reading

Every file below was read in full at argument level, including the proofs and the included figure source; references were inspected as bibliography, not treated as independently read papers. The root for section filenames is `build/source/sections/`. Full SHA-256 values, byte sizes, and line counts are recorded in `evidence/input-hashes.json`. All 22 manuscript snapshot files, including the README, match the stored SHA-256 manifest. This verifies local snapshot consistency, not a fresh network attestation of every commit-to-tree association.

File	Lines	Reading focus
`main.tex` (source root)	147	Assembly, theorem conventions, notation.
`introduction.tex`	175	Target, literature boundaries, proof claims.
`preliminaries.tex`	177	Packing exchange, maximal parts, inverse order.
`hierarchy.tex`	61	Local connectivity and full shortcut quantifiers.
`extraction.tex`	272	Simultaneous response, dyadic counts, compact limit.
`sparsification.tex`	310	Positive isotropy, ranks, split and packing floors.
`iteration.tex`	91	Geometric decay, product bound, final tree.
`mss.tex`	510	Stability, interlacing, partial fractions, barriers.
`fixedpoint.tex`	189	Subdivision, face incidence, Sperner, projection.
`hierarchy_proof.tex`	160	Parameter choices and exponent/logarithm budget.
`hierarchy_dual.tex`	335	Slater dual, singular limits, binary embeddings.
`hierarchy_bags.tex`	337	Truncation, homogeneous bins, two bag types.
`hierarchy_bags_spectral.tex`	211	Tail/rank obstruction and radius mass.
`hierarchy_compact.tex`	326	Shell capacity, reservations, edge-length charging.
`hierarchy_assigned.tex`	41	Assembly and assigned-bag statement.
`hierarchy_assigned_preprocess.tex`	308	Conflicts, labels, excluded descendants, retained mass.
`hierarchy_assigned_order.tex`	106	Ordered bag construction and charging lemma.
`hierarchy_assigned_construction.tex`	457	All three operations, complementary domains, paths.
`hierarchy_assigned_postprocess.tex`	154	Allocations, phase shrink, final energy inequality.
`figures/packing_hierarchy.tex` (source root)	77	Threshold-partition figure and labels.
`references.bib` (source root)	253	Bibliographic identities and external inputs.

The section and equation locators throughout the report make the argument dependencies explicit. A count of files alone would not resolve, for example, the distinction between $P(A)$ and $O(A)$, the domain split in operation 3, or the finite-union closure at extraction's endpoint.

8.2 Private originals and routed theorem passages

The direct private report `res-math/TR-item-224-astra.tex`, 673 lines, was read in full. Its fractional theorem, dependence caveat, coarea and spectral comparisons, full parity transform, positive-domination theorem, and stability discussion were rechecked. Its historical test totals were not replayed and are not counted as this report's tests.

The actual private sources TR190 (393 lines), TR248 (1170 lines), and TR250 (1080 lines) were also read in full, not substituted by their corpus map summaries. They supply, respectively, the graph-spectral filter/frame branch, finite-model perturbation margins, and compatible operator/metric transport. The corrections to TR190 above concern its unit-triangular and exact

heat-locality assertions; the valid spectral branch is retained. These reports' unrelated applications and computational claims were not replayed merely by reading them.

The three corpus maps were used as routing indexes. The actual `RamCalBooks/volumes/volume21/chapters/ch01d-physical-metrics.tex` fragment was read in full (210 lines), including the Gram residual and off-model component. The actual `RamMathUnsolved/TR-meta1-math.tex` passage at lines 160–281 was read for `thm:transport`, `prop:quotient`, and `cor:pd`; the entire 784-line manuscript was not read. These passages corroborate the physical-metric convention; the present certificate proofs are given explicitly above. No unconsulted private bibliography entry is counted as read. Full local paths and hashes are retained in the confidential evidence directory rather than exposed through a public query or upload.

The applicability row for item 320 and its private locators were inspected without modification. The extraction/fixed-point audit was read in full and integrated only within its stated hierarchy-conditional scope. The supplemental resistance note was read in full and its proof reproduced above. Neither note is an independent global verification of the hierarchy.

8.3 Public refresh and formalization boundary

Public-primary retrievals on October 7 used only public mathematical titles and terms. The inspected passages include AOG's locally connected hierarchy theorem and overview; MSS Theorem 1.4; Harvey–Olver Appendix F's halving claims; Klein–Olver's abstract; Klein–Olver–Yeoh's Theorems 1–2; and Kathuria version 2, Theorem 10.2, its reduction, and Section 10.1. The Nash–Williams and Tutte identities are also checked against the manuscript bibliography; the packing argument needed here is read in the manuscript itself. These are targeted literature checks, not a claim to have newly read every external paper in full.

The pinned formalization catalogue lists `OAI.StrongThinTree.strongThinTree`, file `OAI/Combinatorics/ThinTrees/Main.lean`, with comparator configuration `ComparatorChallenges/StrongThinTree.json`. The pinned public main file (66 lines) and comparator (14 lines) were inspected. The main file imports the iteration module and delegates to the multigraph packing thinness theorem; it handles $k \geq 2$ through $r = \lfloor k/2 \rfloor$ and treats $k = 1$ separately. The comparator permits `propext`, `Quot.sound`, and `Classical.choice`, and has `enable_nanoda:false`. The pinned challenge specification was also read in full: it defines a finite indexed loopless multigraph, cut-based connectivity, and a spanning tree as an edge-minimal connected edge set. Its `MainStatement` has one universal positive C, every integer $k \geq 1$, every $n \geq 2$, and one tree satisfying all proper-cut bounds. The specification's `sorry` is a challenge placeholder, not an identified gap in the separate solution file. The import closure, compiler environment, and proof kernel were not replayed. This is catalogue and interface inspection only, with no Lean-verification claim.

8.4 Executed tests and reproducibility limits

The report's own checker is `evidence/check_m320.py`. It uses exact integers and rational `Fraction` arithmetic, with no random seed, floating-point eigensolver, external solver, or theorem prover. Its output `evidence/test-results.json` records assertion groups and inputs. The final successful run has 12,184 assertions, including 44 source-hash and byte-size checks. Its parity test covers 145 labeled trees and 1,997 tree/cut pairs. These counts are this report's own execution only; separately supplied transfer-test totals are not added to them. The tested families are all nontrivial cuts in 24 path-multigraph instances ($2 \leq \ell \leq 9$ and $M \in \{2, \ell, \ell^2\}$, counted with repeated parameter choices); every labeled tree through five vertices via Pruefer codes and every cut mask in a specified positive-multiplicity complete graph; the seven K_4 cut-coordinate identities and rational matrix certificates;

all nontrivial partitions of the doubled K_4 packing example; 113 exact iteration parameter choices $r = m^{16}$, $16 \le m \le 128$; and finite residual and response-budget cases.

The K_4 example was corrected during checking: its maximum is $3/4$, not the initially drafted $2/3$, because the alternating cut crosses all three path edges. The final report and checker agree. This illustrates a useful role of finite tests without promoting them into a universal proof. The checker does not enumerate arbitrary hierarchy arrangements, construct a fixed-point response certificate, run the original private experiments, or validate every precondition of an unexecuted numerical method.

Compilation and all-page rendering records accompany the final source and PDF. The native editor request was queued, but its compiler reported "Unable to find standard directories for platform." Native compilation is therefore unverified; PDF export uses the existing MiKTeX helper and Poppler, without installation. Frozen output hashes are stored separately so the report need not contain its own self-referential hash. Originals, catalogues, and other reports are outside this report's write scope.

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